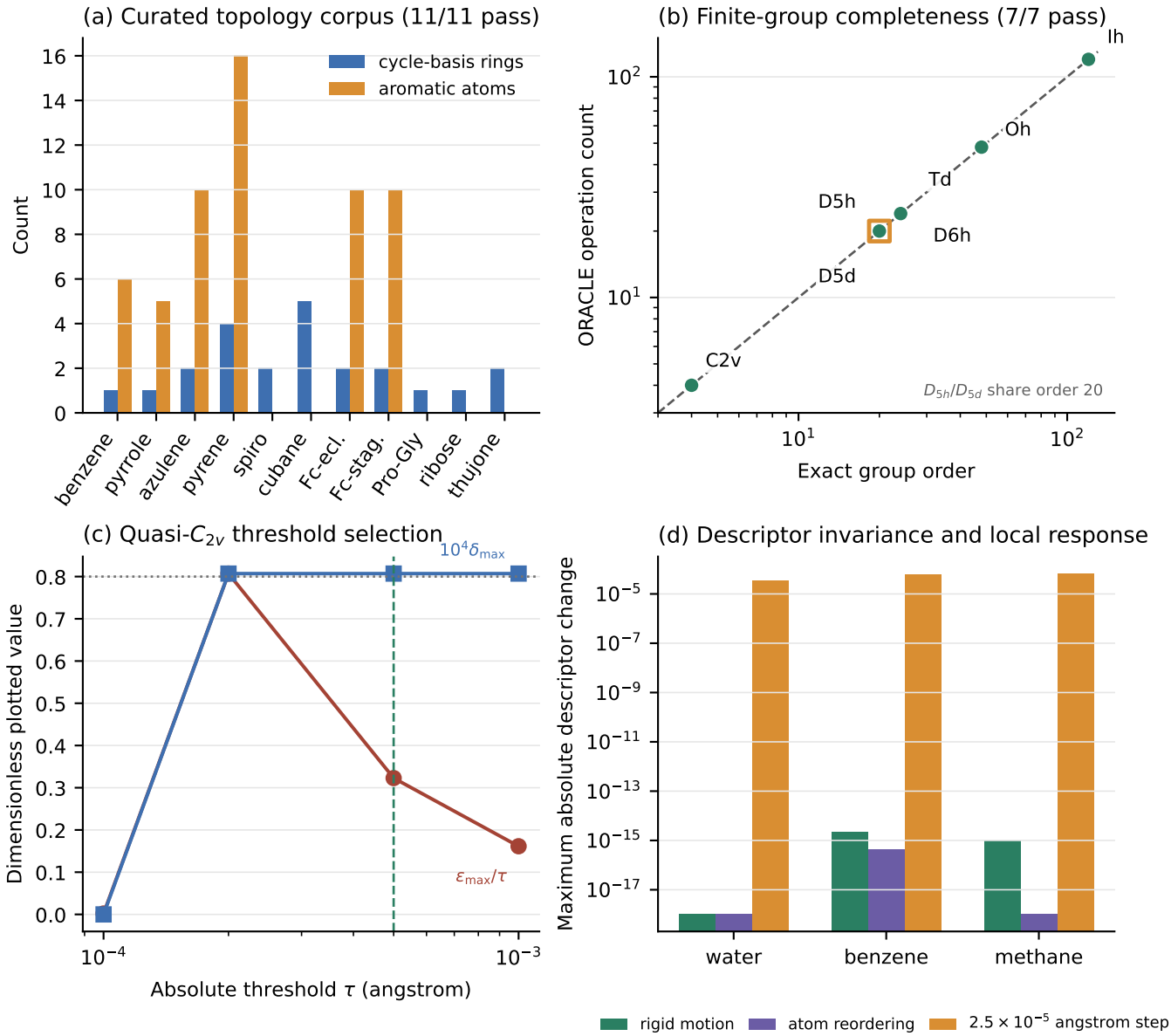
(a) Curated topology corpus (11/11 pass)
cycle-basis rings
aromatic atoms
Count
benzene
pyrrole
azulene
pyrene
spiro
cubane
Fc-ecl.
Fc-stag.
Pro-Gly
ribose
thujone
(b) Finite-group completeness (7/7 pass)
ORACLE operation count
Exact group order
C2v
D5d
D5h
D6h
Td
Oh
Ih
D_{5h}/D_{5d} share order 20
(c) Quasi-C_{2v} threshold selection
$10^4\delta_{max}$
ε_{max}/τ
Dimensionless plotted value
Absolute threshold τ (angstrom)
(d) Descriptor invariance and local response
Maximum absolute descriptor change
water
benzene
methane
rigid motion
atom reordering
2.5×10^{-5} angstrom step